Process Characterization Report

A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) — PCR-009

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCR-009
Document class	Process Characterization Report
Title	Process Characterization Report: Small-Virus Retentive Filtration (Step 9)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; CPP, critical process parameter; Cpk, one-sided process capability index; CQA, critical quality attribute; %CV, coefficient of variation; ICH, International Council for Harmonisation; LOQ, limit of quantitation; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; SOP, standard operating procedure; TCID50, median tissue culture infective dose; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Small-virus retentive filtration removes non-enveloped virus from the anion exchange flow-through pool by size exclusion, and it is the largest single contributor to the parvovirus clearance claimed for A-Mab drug substance. The step is operated in single pass across a dedicated retentive membrane, at a controlled pressure and to a defined volumetric load. This report presents the Stage 1 characterization executed against PCP-009 on a qualified small-scale spiking model. The work follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012) and the viral safety evaluation framework of ICH Q5A(R2) (International Council for Harmonisation 2023a). The process model, the criticality framework and the risk-ranking method are taken from the A-Mab case study (CMC Biotech Working Group 2009).

Both parameters carried into the study, filtration volume and filtration pressure, are classified as well-controlled critical process parameters. The step has no critical process parameter and no key process parameter. Three responses were measured across a screening design of 7 runs and a response-surface design of 12 runs: the MVM log reduction factor, the XMuLV log reduction factor and step yield. The volumetric load is the only parameter with a demonstrated effect on any response. The screening design estimates a reduction in MVM clearance of 1.61 log across the characterized load range ($p = 0.027$), and the response-surface model refines that estimate to 1.73 log.

The response-surface model for MVM clearance is significant and adequate ($R^2 = 0.947$, adjusted $R^2 = 0.903$), and it is the predictive model on which the operating region rests. Neither the XMuLV model nor the step yield model was significant over the ranges studied, so no predictive model is claimed for either. Those two null results are reported as evidence of robustness and are carried into the knowledge space (see §5.3). At the worst-case corner of the characterized region the model predicts an MVM log reduction of 4.48, against a required step contribution of 3.89 log.

At commercial scale both viral-clearance attributes clear their acceptance criteria with margin. The cumulative parvovirus clearance has a one-sided capability index of 1.51 and sits 1.78 log above its criterion; the enveloped-virus attribute has a capability index of 1.53. Parvovirus clearance is the tighter of the two, and this step supplies 5.32 of the 10.03 cumulative log. Two deviations were recorded during execution. Both were investigated, both were dispositioned as retained, and neither altered a parameter classification or the operating region (see §13).

The step does not deliver viral safety on its own. Parvovirus clearance is shared with anion exchange (PCR-008), enveloped-virus clearance is shared with the low-pH hold (PCR-006) and with anion exchange, and the cumulative claim is consolidated in PCMR-001. No clearance is claimed for viruses other than the two model viruses used here, and the capability reported above is an estimate from qualified scale-down models that Stage 2 confirms at commercial scale.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a fed-batch mammalian cell culture and purified by the Novacyte platform train. The purification sequence is Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and UF/DF. Small-virus retentive filtration is Step 9 of the drug substance process, and it sits between the final chromatographic polish and formulation. It receives the anion exchange flow-through pool as its load (the composition of that pool is set by the design space of Step 8, and is reported in PCR-008) and delivers the filtrate to the UF/DF step.

The step removes virus by size exclusion. The membrane retains particles above its pore-size distribution while the antibody monomer passes, so clearance depends on the physical dimensions of the virus and on how much material the retentive layer has already seen. This mechanism is orthogonal to the two upstream viral steps. Low-pH inactivation destroys the lipid envelope of enveloped virus (PCR-006), and anion exchange removes virus by charge-based binding in flow-through mode (PCR-008). A filtration step that works on size alone therefore adds clearance that does not share a failure mode with either of them. That is the basis of the modular claim made in ICH Q5A(R2) (International Council for Harmonisation 2023a).

The step forms no product quality attribute of its own, and it contributes to two attributes of the drug substance: the cumulative clearance of minute virus of mice (MVM) as the parvovirus model, and the cumulative clearance of xenotropic murine leukaemia virus (XMuLV) as the enveloped model. MVM is the harder of the two to remove, because a parvovirus is small enough that its retention is set by the tail of the pore-size distribution. The filter is single use, so no resin or membrane life-cycle claim arises at this step. Performance of the nominal batch is given in Table 1.1.

Table 1.1: Nominal-batch performance of the small-virus retentive filtration step.

Metric	Nominal batch
Step yield (fraction of input product mass)	0.984
Product mass out (g)	56,232
XMuLV log reduction (log10)	5.8246
MVM log reduction (log10)	5.3221

Step yield is high and both log reduction factors sit well above the contribution the step is required to make (that contribution is derived in §7). The nominal batch is the at-set-point condition, so these values are the reference against which the characterization ranges were explored.

1.2 Objectives

The objectives of this characterization were:

(i) to quantify the relationship between each studied process parameter and the viral clearance and process performance responses of the step;
(ii) to define the operating region over which the required clearance contribution is met;
(iii) to establish proven acceptable ranges for each parameter against each governed quality attribute;
(iv) to classify each parameter according to its demonstrated effect and the risk of operating outside the region; and
(v) to justify the ranges taken forward into Stage 2 and the controls carried into the control strategy.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the FDA lifecycle guidance (U.S. Food and Drug Administration 2011), and this report is the study record for that stage. The development approach is the enhanced approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), in which prior knowledge and a formal risk assessment set the scope of experimentation and the outcome is a control strategy supported by process understanding. Criticality is treated as a continuum in the sense of ICH Q9 (International Council for Harmonisation 2023b), so a quality attribute is assigned a level of criticality and the attributes at high and very high criticality are called critical.

Viral safety is evaluated under ICH Q5A(R2) (International Council for Harmonisation 2023a), a framework that requires clearance to be demonstrated on a qualified scale-down model with deliberately spiked virus, because live virus cannot be introduced into a manufacturing facility. It also requires that the steps credited in a cumulative claim remove virus by mechanisms that are distinct, so that one failure cannot remove the whole margin, and both of those requirements shape the study reported here. The process model, the CQA register and the risk-ranking method follow the A-Mab case study (CMC Biotech Working Group 2009).

1.4 Related documents and controlled procedures

This report is one of the per-step reports that roll up into the master report (PCMR-001), and its place in the package is fixed by three others: its scope was set by the pre-characterization risk assessment (RA-001), its method was fixed in advance by the paired plan (PCP-009), and its conclusions feed the campaign-level control strategy. The package is listed in Table 1.2.

Table 1.2: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-009	Process Characterization Plan (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and validated analytical methods that govern the work are listed in Table 1.3. Operation of the step, the pre-use flush and the post-use filter integrity test are covered by SOP-2012. The scale-down model was qualified under SOP-1001, the designs were built and analysed under SOP-1002, and the classification decisions in §9 follow SOP-4001.

Table 1.3: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Small-virus retentive filtration is a mature platform step at Novacyte, and the prior knowledge available for it is substantial. The same filter type and the same operating mode have been used for the three previously licensed antibodies on the platform, X-Mab, Y-Mab and Z-Mab, and the step has been characterized for each of them. The mechanism of removal is identical across those products, because retention is set by the membrane and by the size of the virus. The antibody takes no part in it. Prior knowledge therefore transfers to A-Mab, and the purpose of this study is to confirm and bound the platform behaviour for this molecule.

Two expectations followed from that experience and framed the design. The first is that MVM clearance falls as the cumulative volumetric load rises. Material deposited on and within the retentive layer changes the effective pore-size distribution as the filtration proceeds, and a particle as small as a parvovirus is the first to benefit from that change. Enveloped virus is large enough that the same shift does not reach it, so XMuLV clearance was expected to be flat across the load range. The second expectation is that pressure has little influence on retention. Pressure sets the flux, and with it the processing time, but a size-based retention mechanism is not driven by convective velocity over the range a filter of this type is qualified for.

These expectations set the ranges as much as they set the factor list. The load range was extended well beyond the routine ceiling in order to find where clearance stops being sufficient, and the pressure range was extended to both operating limits of the filter so that a null result would be a bounded null result. Neither expectation was treated as established for A-Mab before the study, and both are tested in §5.

2.2 Quality attributes in scope

The step sets no quality attribute of its own, and it governs the two viral-clearance attributes of the drug substance. Both are listed in Table 2.1 with their acceptance criteria and their assigned criticality.

Table 2.1: Quality attributes governed by the step, from the drug-substance CQA register.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

Both attributes are of very high criticality, which places them above the threshold at which the A-Mab framework calls an attribute critical (CMC Biotech Working Group 2009). Both acceptance criteria are cumulative one-sided floors on the whole purification train, so neither can be assessed against this step alone, and the criterion this step must meet is the cumulative requirement less the clearance credited to the other steps (the back-calculation is stated in §7). Product quality attributes formed upstream, such as the glycan and charge variant distributions, are not affected by a size-based filtration step and were not measured here. Aggregate was not measured here either. No change in aggregate is expected across a filtration that separates by size and passes the antibody monomer, and aggregate is reduced and reported at the cation exchange step (PCR-007).

2.3 Risk-based prioritization and parameter selection

The scope of this study was fixed by RA-001. That assessment ranked every parameter of the step before any characterization data existed, and it scored the severity of the potential effect on a CQA together with the likelihood of occurrence and the ability to detect the failure in routine manufacture. Its output is a study assignment, not a classification. The rows for this step are reproduced in Table 2.2.

Table 2.2: Pre-characterization risk assessment for the step (RA-001), with the assigned study type.

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	Initial RPN	Assigned study	Priority
Filtration volume (load)	Volumetric load above 105 L/m2	Reduced MVM log-reduction (parvovirus breakthrough risk)	10	7	10	700	multivariate DoE	High
Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load	10	7	10	700	multivariate DoE	High

Both parameters scored at the top of the range and both were assigned to a multivariate design. The severity term is the same for each, because the consequence of interest is a parvovirus breakthrough and that consequence is scored at the maximum. What separates the two rows is the reason each parameter could reach that consequence. Volumetric load reaches it directly, by consuming the retentive capacity of the membrane, whereas pressure reaches it indirectly, by moving the filter away from the operating window in which its retention has been established by the vendor and by the platform.

No parameter at this step was assigned to univariate assessment and none was excluded from study. That follows from the short parameter list the step has. Membrane lot, pre-use flush volume and the post-use integrity test are attributes of the filter and of the procedure, controlled under SOP-2012, and they are not process parameters in the sense of RA-001. The load material is the anion exchange flow-through pool, whose composition is controlled by the design space of Step 8 (PCP-008 and PCR-008), so feed variation was not introduced as a factor here.

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were executed on a qualified scale-down model of the commercial filtration step, operated under SOP-2012 and qualified under SOP-1001. The model uses the same membrane chemistry and the same retentive layer as the commercial filter, in a device of smaller area from the same vendor family. Scaling is by filter area. Volumetric load is expressed per unit membrane area, so a load in litres per square metre carries the same meaning at both scales, and pressure is a scale-independent set-point that is matched directly (the qualification protocol is SOP-1001). The model thus reproduces the two quantities that the mechanism of retention depends on.

Qualification compared the scale-down device with commercial-scale data at the target condition, using flux decay across the filtration, filtrate quality and step yield as the performance measures. No difference of practical consequence was found between the scales, and the qualification record is held under SOP-1001. This is the warrant for reading the operating ranges established here as ranges for the commercial step.

Viral clearance cannot be measured at commercial scale, because live virus is not introduced into a manufacturing facility. Clearance is therefore measured on a small-scale spiking model, as ICH Q5A(R2) requires (International Council for Harmonisation 2023a). The load material for each run was the anion exchange flow-through pool, spiked with the model virus to a titre high enough that a clearance of several logs can still be quantified above the detection limit of the assay. Spiking adds material to the load, so the spike volume was held to a small fraction of the load volume and the spiked load was clarified before filtration. Both precautions are part of the qualified method and both are recorded per run.

The centre-point replicates of each design were executed as independent runs on independent filter devices, and not as repeat assays of one filtrate. They carry the whole run-to-run variability of the model, and their scatter is the pure-error term used in the lack-of-fit assessment (§5.3).

3.2 Operation

The commercial step is a single-pass filtration at constant pressure. The filter is flushed and pressure-tested before use, the load is delivered from the anion exchange pool at the target pressure, and filtration continues until the batch volume has passed or the volumetric load limit is reached. A post-use integrity test is performed on every filter, and a filter that fails it invalidates the batch. All of this is specified in SOP-2012 and was followed in the scale-down runs.

Two quantities are set by the operator. The volumetric load is fixed by the batch volume and the installed membrane area, and it is the quantity the operator manages when the filter area is selected for a batch. The filtration pressure is set at the start of the filtration and held, and everything else at the step is either an attribute of the filter, which is controlled by the vendor specification and by incoming inspection, or an attribute of the load, which is controlled by the preceding step (PCR-008).

3.3 Analytical methods

Two validated infectivity assays supply the responses of the study. MVM titre is determined by AMV-3018, which combines a TCID50 endpoint assay with a qPCR confirmation, and XMuLV titre is determined by AMV-3017, a TCID50 endpoint assay. The log reduction factor for each run is the difference between the log titre of the spiked load and the log titre of the filtrate, both corrected for the volumes involved. Step yield is a protein mass balance across the filter, computed from the in-process concentration and volume records required by SOP-2012, so it is not an assay result and has no separate method validation. Table 3.1 states which assay supplies which response.

Table 3.1: Validated analytical methods and the response each supplies.

Method	Title	Analyte	Reported response
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Minute virus of mice (MVM)	MVM LRF
AMV-3017	XMuLV Infectivity Titre (TCID50)	Xenotropic murine leukaemia virus (XMuLV)	XMuLV LRF

An infectivity assay is considerably less precise than a chromatographic purity assay, and that matters for how the results in §5 are read. The validated intermediate precision of each method, its limit of quantitation and the share of observed variance attributable to the method are given in Appendix D. Where an observed difference in log reduction is of the same order as the assay precision, it is reported as such and is not interpreted as a process effect.

3.4 Sampling plan

Each run produced a spiked load sample and a filtrate sample, both assayed for the model virus of that run, plus load and filtrate samples for the protein mass balance. Samples were drawn under SOP-2012 and assayed within the validated hold conditions of the method.

The screening design carried 3 centre-point replicates and the response-surface design carried 4. Centre points were spread through the run order and were not executed consecutively, so that a drift in the assay or in the load material would appear as scatter between them. Run order within each design was randomized.

3.5 Statistical methods

Factors were coded so that the midpoint of each characterization range is zero, giving coded levels of −1 / +1 at the two edges. Coding puts the two parameters on a common scale, so effects can be compared directly, and it is the form in which every model in this report was fitted.

The screening data were fitted by ordinary least squares with both main effects and their interaction. An effect is reported as twice the coded coefficient, which is the change in the response across the full characterization range of that factor. The response-surface data were fitted with the full quadratic model, adding a squared term for each factor. Significance is assessed at alpha = 0.05 throughout.

Model adequacy is judged on four measures taken together. R^2 and adjusted R^2 describe the fit, predicted R^2 from the PRESS statistic describes how the model performs on data it has not seen, the overall F test asks whether the model explains more than the residual, and the analysis of variance partitions the residual into lack of fit and pure error so that the model form can be tested against the replicate scatter. A response for which no term is active is reported as robust over the ranges studied, and no predictive model is claimed for it.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter drawn inside its normal operating range, and is reported as a one-sided Cpk against the relevant acceptance criterion. The proven acceptable ranges in §7 use the same fitted response-surface model, once with the other parameter fixed and once with it varying inside its normal operating range.

4 Study design

4.1 Factors, ranges and the knowledge space

The design resolves the effect of each parameter on each response and then defines the region in which the required clearance is delivered. Both parameters of the step were studied, and their set-points, normal operating ranges, characterization ranges and final classifications are given in Table 4.1. The coded letters used in the effect and coefficient tables are given in Table 4.2.

Table 4.1: Process parameters of the step, with normal operating range, characterization range and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Filtration volume (load)	L/m2	90	50–105	50–140	WC-CPP	multivariate
Filtration pressure	psi	13	10–20	8–30	WC-CPP	multivariate

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m2	screening + RSM
B	Filtration pressure	psi	screening + RSM

The characterization ranges are wider than the normal operating ranges on both axes, and the two widenings were chosen for different reasons. Volumetric load runs from 50–140 L/m², against a normal operating range of 50–105 L/m². The upper edge is well above the routine ceiling, because prior knowledge says clearance declines with load, and the point of the study is to establish how far that decline can go before the step stops meeting its required contribution (§7). A range that stopped at the routine ceiling could not have answered that question. Filtration pressure runs from 8–30 psi against a normal operating range of 10–20 psi, and here the edges are the operating limits of the filter itself. Testing to the vendor limits is what turns an expected null result into a bounded one.

The characterization ranges together define the knowledge space of the step. The design space established in §6 is a subset of that space, and the normal operating ranges sit inside the design space. That nesting is the structure ICH Q8 describes, and it is what allows a parameter excursion to be assessed against evidence rather than against judgement (International Council for Harmonisation 2009).

4.2 Screening design

The screening design is a two-level full factorial in the two parameters, augmented with 3 centre points, giving 7 runs in total. A full factorial in two factors estimates both main effects and the interaction without confounding, so the factorial part is exactly saturated. The residual degrees of freedom come entirely from the centre points, and those runs also supply the replicate scatter and the test for curvature. The design matrix and the measured responses are in Appendix A.

The purpose of this stage is to identify which parameters have an effect worth modelling. It is not to quantify that effect precisely, and this design could not do so. With three model terms and 7 runs the residual carries few degrees of freedom, so the standard errors are wide and the test has limited power.

4.3 Response-surface design

The response-surface design is a face-centred central composite design in the same two parameters, with 4 centre points and 12 runs. The axial points are placed on the faces of the design cube, so no run leaves the characterization range. Adding the axial points allows a squared term for each factor, so the model can describe curvature in the clearance surface. The design matrix and the measured responses are in Appendix B.

Both parameters appear in both designs, because there are only two of them. In this study the response-surface stage is not a reduction of the factor set, as it usually is. It augments the same factor set with the runs needed to fit curvature and to obtain a model that can be used for prediction. The division of labour still holds. The screening design identifies which parameters are active, and the response-surface model is the predictive model and the basis of the design space (§6).

4.4 Univariate assessment

No parameter of this step was assessed univariately. RA-001 assigned both parameters to the multivariate design, and there was no third parameter left over to assess one at a time. Attributes of the filter and of the procedure that could in principle affect retention, such as membrane lot and pre-use flush volume, are controlled under SOP-2012 and by the vendor specification, and they are treated as controlled inputs. One of them became the subject of a deviation during execution, and that is reported in §13.2.

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility of the scale-down model is adequate for the effects the study set out to resolve, and the two log reduction factors are the least precise of the three responses. Table 5.1 gives the centre-point mean, standard deviation and coefficient of variation for each response in the response-surface design, and Table 5.2 gives the same for the screening design.

Table 5.1: Centre-point reproducibility in the response-surface design.

Response	n	Mean	SD	%CV
MVM LRF ($\log_{10}$)	4	5.54	0.116	2.1
XMuLV LRF ($\log_{10}$)	4	6.13	0.176	2.87
Step yield	4	0.983	0.00839	0.854

Table 5.2: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
MVM LRF ($\log_{10}$)	3	5.44	0.488	8.99
XMuLV LRF ($\log_{10}$)	3	6.3	0.147	2.34
Step yield	3	0.986	0.00767	0.777

Step yield reproduces most tightly, at a coefficient of variation of 0.85 % in the response-surface design. A mass balance across a filter that plugs very little should reproduce about that well. The two log reduction factors are less precise, at 2.10 % for MVM and 2.87 % for XMuLV, and the enveloped-virus response is the noisier of the two in this design. In absolute terms the MVM centre-point standard deviation is 0.116 log, against a validated intermediate precision of 0.32 log for AMV-3018. The observed run-to-run scatter is below the intermediate precision of the assay, so the reproducibility of the model cannot be resolved separately from the method at this sample size. The validation of AMV-3018 attributes a fraction of 0.3 of observed variance to the method (Appendix D).

The screening centre points are noisier for MVM, at 8.99 %, with a standard deviation of 0.488 log. That scatter is of the same order as the assay precision quoted above, and it rests on only 3 replicates, so it is read here as assay variability in a small sample. It is the reason the screening standard errors in §5.2 are wide. The response-surface design carries more centre points and shows the tighter figure.

5.2 Screening: factor effects

The screening stage identified volumetric load as the only active parameter, and it identified it on one response only. Model-level statistics for the three screening fits are given in Table 5.3.

Table 5.3: Screening model adequacy, all responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
MVM LRF ($\log_{10}$)	7	0.848	0.695	0.769	5.56	0.0962	0.4
XMuLV LRF ($\log_{10}$)	7	0.714	0.428	0.364	2.5	0.236	0.122
Step yield	7	0.709	0.418	-2.23	2.43	0.242	0.0073

None of the three screening models is significant overall at alpha = 0.05, including the MVM model, whose overall p-value is 0.096. The screening design has 7 runs against three model terms, so the residual carries few degrees of freedom and the overall F test is weak by construction. What the MVM model does show is a single large term against a small residual, and the finding rests on that term. The step yield model has a negative predicted R^2, which means it predicts a held-out run worse than the response mean does, and no conclusion is drawn from any of these three fits beyond which terms are worth carrying into the response-surface stage.

Table 5.4: Screening effect estimates for MVM log reduction.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-1.61	-0.807	0.2	-4.04	0.0273	*
B	-0.246	-0.123	0.2	-0.617	0.581	
AB	0.0192	0.00962	0.2	0.0482	0.965	

Volumetric load reduces MVM clearance by 1.61 log across the characterized load range, and it is the only term in Table 5.4 that reaches significance (p = 0.027). The sign is the one prior knowledge predicted, since clearance falls as more material passes the membrane. Pressure has an effect of 0.246 log in the same direction. That estimate is smaller than the centre-point scatter of the same design and does not approach significance. The interaction term is smaller still, and its estimate of 0.019 log is two orders of magnitude below the load effect.

Table 5.5: Screening effect estimates for XMuLV log reduction.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.262	0.131	0.0608	2.16	0.12	
AB	0.188	0.0939	0.0608	1.54	0.22	
A	-0.0821	-0.041	0.0608	-0.675	0.548	

Enveloped-virus clearance shows no significant dependence on either parameter (Table 5.5). The largest term is pressure, at 0.262 log across the pressure range, and it does not reach significance. Load, which drives the MVM response, is the smallest of the three terms here. The contrast between the two model viruses is exactly the contrast prior knowledge expected, and §5.4 takes it up. Step yield shows no significant term either, and its full effect table is in Appendix C.

The screening stage carried one active term forward. The magnitudes above are estimates from a design built to detect effects, and they are refined by the response-surface model (§5.3), which is the model used for prediction.

5.3 Response-surface models

One of the three response-surface models is adequate and predictive, and it is the model for the response that governs the step. Model-level statistics for all three are given in Table 5.6.

Table 5.6: Response-surface model adequacy, all responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
MVM LRF ($\log_{10}$)	12	0.947	0.903	0.527	21.4	0.000921	0.211
XMuLV LRF ($\log_{10}$)	12	0.506	0.0949	-1.14	1.23	0.398	0.158
Step yield	12	0.26	-0.356	-1.56	0.422	0.819	0.00685

The MVM model explains $R^2 = 0.947$ of the variation with an adjusted R^2 of 0.903, and the overall F test gives $p = 0.0009$. Its predicted R^2 is 0.527, which is materially lower than the fitted R^2 and is the honest measure of how the model behaves on a run it has not seen, so predictions near the edge of the region should be read with that gap in mind. The residual standard error is 0.211 log, of the same order as the assay precision quoted in §5.1.

Neither of the other two models is significant. The XMuLV model gives $p = 0.398$ with an adjusted R^2 of 0.095, and the step yield model gives $p = 0.819$. Both have negative predicted R^2. These are null results and they are reported as such. No predictive model is claimed for enveloped-virus clearance or for step yield at this step, and neither response is used to constrain the operating region. Both remain in the knowledge space as evidence that the step is insensitive to load and to pressure over the ranges studied. Their coefficient tables and analyses of variance are in Appendix C.

Table 5.7: Response-surface coefficients for MVM log reduction (coded factors).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.52	0.0962	57.4	1.87e-09	***
A	-0.866	0.086	-10.1	5.57e-05	***
B	-0.0341	0.086	-0.396	0.706	
AB	0.028	0.105	0.266	0.799	
A^2	0.136	0.129	1.05	0.333	
B^2	-0.301	0.129	-2.34	0.0582	

The coefficient table confirms the screening result and adds the curvature terms (Table 5.7). The load coefficient is the only term significant at alpha = 0.05, and its standard error is small enough that the effect is resolved rather than merely detected. The squared term in pressure has a p-value of 0.058, close to the threshold without crossing it. It describes a shallow maximum in clearance near the middle of the pressure range, and its practical size is small. Since it is not significant it is not treated as a mechanism. It is nevertheless retained in the model, for two reasons. Dropping a term this close to the threshold in order to refit a five-term model on 12 runs would overstate the precision of the coefficients that remain. The surface fitted with the term in place is also the more conservative basis for the worst-case corner evaluated in §6. The interaction term is not significant, which means the effect of load on clearance does not depend on the pressure the filter is run at.

Table 5.8: Analysis of variance with lack-of-fit partition for the MVM model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	4.75	5	0.951	21.4	0.000921
Residual	0.266	6	0.0444	nan	nan
Lack of fit	0.226	3	0.0752	5.55	0.0965
Pure error	0.0406	3	0.0135	nan	nan

The analysis of variance in Table 5.8 supports the model form with one qualification. Lack of fit is not significant at alpha = 0.05, with p = 0.096, so the quadratic surface is not rejected against the replicate scatter. The margin is small, and the pure-error estimate rests on 3 degrees of freedom, so this test is not a strong one. Taken with the modest predicted R^2, it means the model should be used to define an operating region (§6) and to predict mean levels inside it, and not to make point predictions at the corners with a claimed precision.

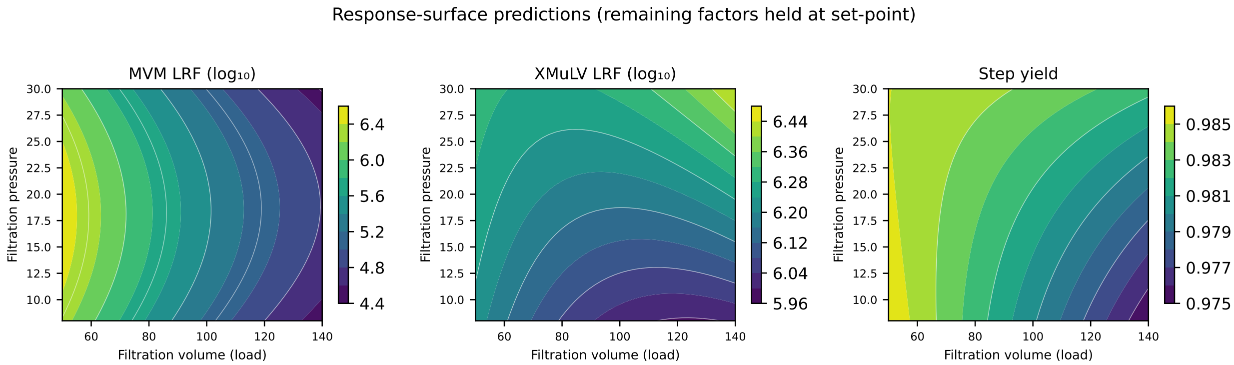

Figure 5.1: Response-surface predictions across the characterized load and pressure ranges, for each of the three measured responses.

The surfaces in Figure 5.1 make the shape of the result visible. MVM clearance falls steadily from left to right across the load axis and is almost flat along the pressure axis, so its contours are close to vertical lines. The XMuLV and step yield panels are drawn from models that are not significant, and the small tilts they show should not be read as effects. They are shown for completeness, and because a flat surface is the visual form of the robustness claim made above.

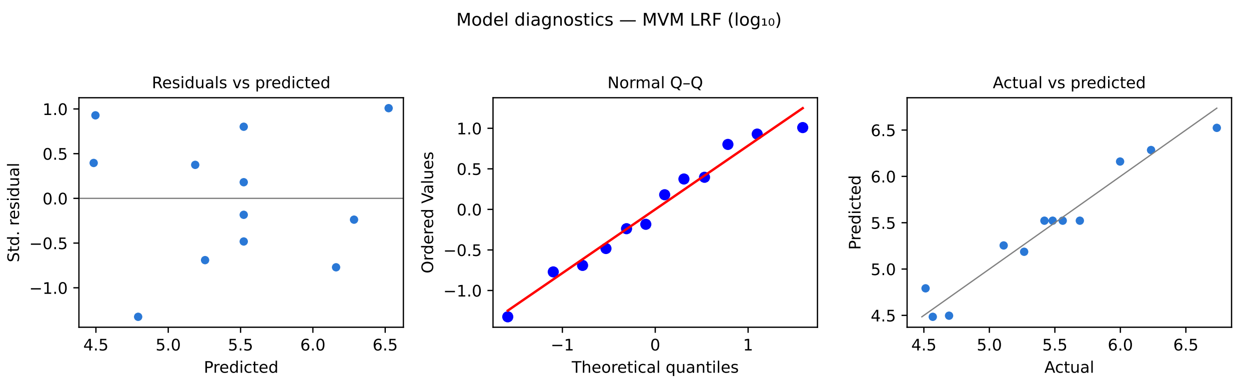

Figure 5.2: Model diagnostics for the MVM response-surface model: standardized residuals against predicted values, normal quantile plot, and actual against predicted.

Diagnostics for the MVM model are given in Figure 5.2. The residuals show no structure against the predicted value, so the variance is stable across the range of clearance the model covers. The normal quantile plot is close to a straight line, with no point far enough from it to suggest an outlier or a failed run. The actual against predicted panel sits on the identity line across the whole range. Nothing in the diagnostics contradicts the model form, which is the most that diagnostics on 12 runs can establish.

5.4 Mechanistic interpretation

The results are consistent with retention by size exclusion on a membrane whose effective pore-size distribution changes as material accumulates. Every finding of the study follows from that one mechanism, and the two model viruses behave differently for the reason that mechanism predicts.

MVM is at the small end of what the membrane can retain. Its retention depends on the tail of the pore-size distribution, so anything that shifts that tail changes clearance. As the volumetric load rises, protein and other load components deposit within the retentive layer, and the passage available to a particle of parvovirus size changes with them. Clearance declines steadily with load, and the load coefficient in Table 5.7 measures that decline. XMuLV is several times larger and is retained with a wide margin over the whole range, so the same shift in the tail of the distribution does not reach it. The flat XMuLV surface is not an absence of information. It is the expected consequence of a size-based mechanism acting on a virus that is comfortably above the retention threshold.

Pressure behaves as expected as well. Over the range tested, pressure sets the flux, and with it the time the filtration takes, and it does not change what the membrane retains. That is why the pressure terms are small on every response and why the interaction with load is negligible. The practical consequence is that the two parameters can be treated as acting independently inside the region, so the operating region is close to a rectangle in the two parameters.

A multivariate design was still the right instrument, because independence is a finding of the study and not an assumption of it, and the interaction term had to be estimated in order to be dismissed. Had the interaction been active, univariate ranges on the two parameters would have described a region the process does not have. The step yield result belongs to the same picture. Yield stays high across the whole region, including at the highest load. The membrane is therefore not plugging to any appreciable degree over the loads tested, and the decline in MVM clearance comes from a change in the retentive layer rather than from gross fouling.

6 Design space

The design space of the step is the whole characterized region, bounded by the load and pressure ranges studied. Within 50–140 L/m² of volumetric load and 8–30 psi of filtration pressure, the fitted response-surface model predicts an MVM log reduction at or above the contribution the step is required to make, and no corner of that region breaches it. The region has one principal plane, load against pressure, because those are the only two parameters of the step.

The worst case in that region is the corner at the highest load and the highest pressure. The model predicts an MVM log reduction of 4.48 there, against a required step contribution of 3.89 log, leaving a margin of 0.59 log. Load is what puts that corner where it is. The shallow curvature term discussed in §5.3 places the pressure minimum at an edge of the pressure range instead of in the middle of it, and the two pressure edges then differ by 0.01 log at the highest load, against a residual standard error of 0.211 log. That difference is well inside the scatter of the model, so load alone sets the worst case, and a reader tracking the limiting condition of this step should track the load axis. At the upper corner of the normal operating ranges the predicted clearance is 5.33 log, a margin of 1.44 log over the requirement, and at the set-point it is 5.55 log.

The operating ranges are nested inside the design space in the way ICH Q8 describes (International Council for Harmonisation 2009). The normal operating range for load, 50–105 L/m², is a strict subset of the characterized range, and the same holds for pressure at 10–20 psi. The knowledge space is wider than both, since the study deliberately explored loads well above the routine ceiling. Movement within the design space is not a regulatory change, though it is still assessed under the pharmaceutical quality system before it is made (International Council for Harmonisation 2008). Movement outside the design space requires change control and, for a viral clearance step, a fresh assessment of whether the clearance credited in the cumulative claim still holds.

Three bounds apply to this design space. It is bounded by the ranges studied, and nothing is claimed for a load above 140 L/m² or a pressure outside 8–30 psi. It is bounded by the model, which predicts mean levels of clearance from 12 runs and has a predicted R^2 of 0.527, so the assurance available at the very edge of the region is lower than at its centre. It is bounded by the scale-down spiking model, which is the only place clearance can be measured, and by the assumption that the load material stays within the composition the anion exchange design space delivers (PCR-008). The design space is also constrained by things the model does not describe. A filter that fails its post-use integrity test invalidates the batch whatever the model predicts, and the vendor operating limits for the device remain in force at both edges of the pressure range.

7 Proven acceptable ranges

The proven acceptable ranges complement the design space by stating, for each governed quality attribute and each parameter separately, the range over which the attribute stays within acceptance. The acceptance basis comes first, because for a viral clearance response it is not the drug substance specification. The specification is a cumulative floor over the whole train, so the criterion for this step is that floor less the clearance credited to the other steps that contribute. Table 7.1 states the resulting criterion for each response.

Table 7.1: Acceptance criterion for each measured response, with its basis.

Response	Quality attribute	Acceptance criterion	Unit	Basis
MVM LRF ($\log_{10}$)	Viral clearance — MVM (parvovirus)	≥ 3.89	$\log_{10}$ (this step)	Cumulative requirement less the clearance credited to the other steps
XMuLV LRF ($\log_{10}$)	Viral clearance — XMuLV (enveloped)	≥ 3.65	$\log_{10}$ (this step)	Cumulative requirement less the clearance credited to the other steps

The step must deliver at least 3.89 log of MVM clearance and at least 3.65 log of XMuLV clearance. Both criteria are well below what the step achieves at its set-point. Step yield maps to no quality attribute and so has no acceptance criterion here; it is a process performance measure and is treated as such.

Two analyses were run for each pairing of attribute and parameter. The first holds the other parameter at its set-point and evaluates the fitted response-surface model across the parameter's characterization range. The second propagates the normal operating range of the other parameter by Monte-Carlo simulation of the fitted model, with 2,000 draws at each grid point and the model residual added, and takes the range over which the predictive interval stays within acceptance. The second is the reproducible default, because it asks whether the range still holds when the rest of the operating point moves as it normally does. The results are in Table 7.2.

Table 7.2: Proven acceptable ranges per governed quality attribute and parameter, from both analyses.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
MVM LRF ($\log_{10}$)	Filtration volume (load)	50–140	L/m2	50–140	50–140
MVM LRF ($\log_{10}$)	Filtration pressure	8–30	psi	8–30	8–30
XMuLV LRF ($\log_{10}$)	Filtration volume (load)	50–140	L/m2	50–140	50–140
XMuLV LRF ($\log_{10}$)	Filtration pressure	8–30	psi	8–30	8–30

Every entry in Table 7.2 spans the full characterization range, and the two analyses agree on every row. The whole studied range of volumetric load is a proven acceptable range for MVM clearance, and it stays a proven acceptable range when pressure varies across its normal operating range. The same holds for pressure, and both parameters behave the same way against the XMuLV attribute. Nothing narrows. There is no binding univariate constraint anywhere in this step, and clearance never falls to its floor inside the region (§5.4).

That result has a consequence for how the ranges should be used. Because no proven acceptable range is narrower than the characterization range, the binding constraint on the operating region is not any univariate range. It is the multivariate corner identified in §6, the point of highest load and highest pressure, where the margin over the requirement is smallest. A reader looking for the limiting condition of this step should look there.

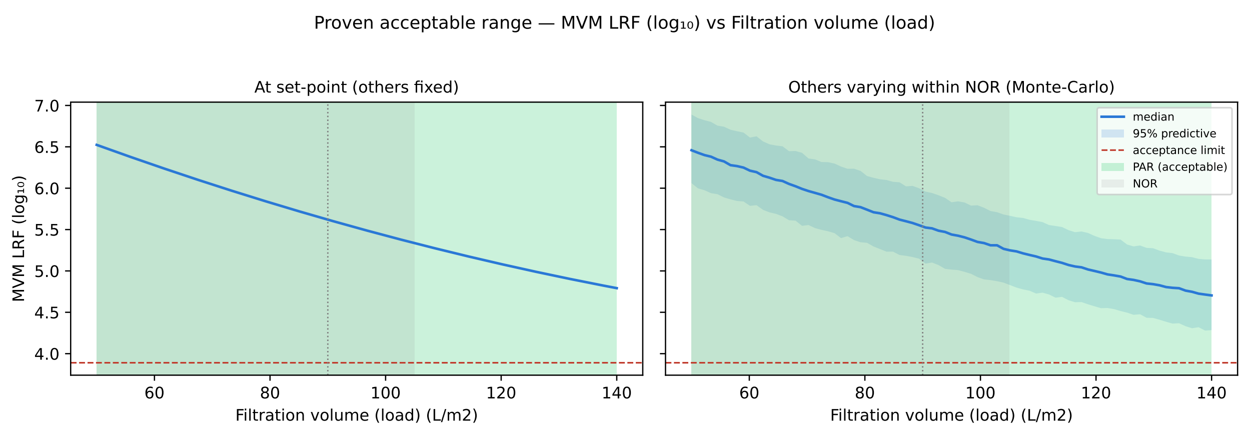

Figure 7.1: Proven acceptable range for MVM log reduction against volumetric load. The acceptance limit is drawn, the acceptable region is shaded green, and the set-point and normal operating range are marked. Left panel: the other parameter at set-point. Right panel: the other parameter varying within its normal operating range.

Figure 7.1 shows the governing pairing of the step, MVM clearance against volumetric load. The predicted clearance declines across the load range in both panels and stays above the acceptance limit throughout, so the shaded acceptable region covers the whole axis. The right panel shows the predictive interval widening around the median once pressure is allowed to vary within its normal operating range, and the lower bound of that interval still clears the limit at the highest load. The set-point sits in the left half of the range, with the normal operating range around it.

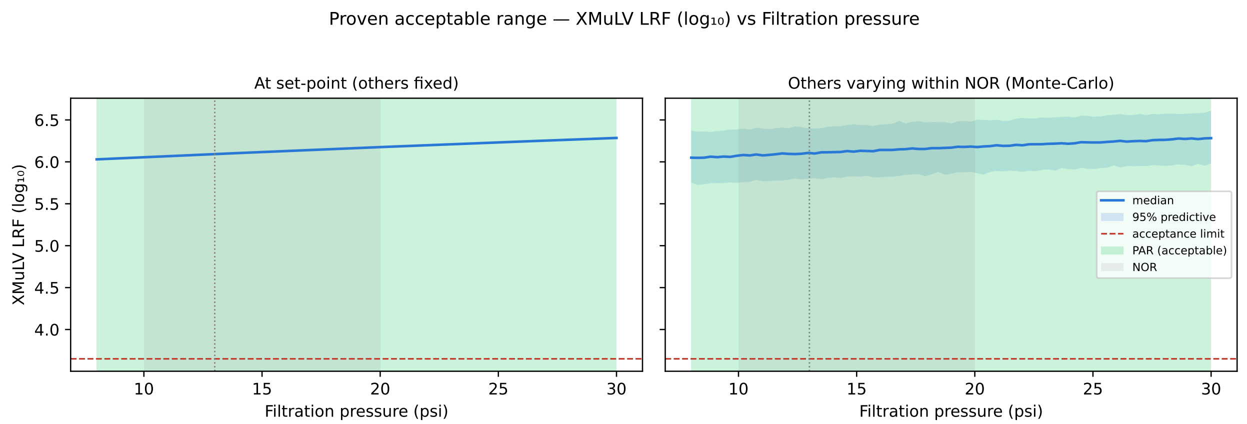

Figure 7.2: Proven acceptable range for XMuLV log reduction against filtration pressure, with the acceptance limit drawn and the acceptable region shaded green.

Figure 7.2 shows the equivalent pairing for enveloped virus, plotted against pressure because that is the parameter with the largest main effect on the XMuLV response. The predicted clearance is essentially flat and sits far above the acceptance limit at every pressure tested. No effect was found, the range over which no effect was found is stated, and the distance to the limit is visible.

Two bounds apply to the ranges in Table 7.2. They are conditional on the fitted response-surface model, so they inherit the predicted R^2 and the residual scatter reported in §5.3, and they extend no further than the characterization ranges over which that model was fitted. They are also conditional on the step's acceptance criterion, a back-calculation from the cumulative requirement. If the clearance credited to anion exchange or to the low-pH hold changed, the criterion for this step would change with it, and the ranges would have to be re-evaluated against the new floor.

8 Process capability and robustness

Both viral-clearance attributes of the drug substance meet their acceptance criteria at commercial scale with margin, and the parvovirus attribute is the tighter of the two. Table 8.1 gives the simulated distribution and the one-sided capability index for each.

Table 8.1: Commercial-scale capability of the viral-clearance attributes the step governs.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

Cumulative parvovirus clearance has a capability index of 1.51 and a simulated mean 1.78 log above its acceptance floor. Cumulative enveloped-virus clearance has a capability index of 1.53 and a mean 2.81 log above its floor. The parvovirus figure is the tighter capability of the two, and it is the one this step contributes most to.

Three bounds apply to this claim. The estimate comes from Monte-Carlo simulation of 2,000 batches on qualified scale-down models, so it is a prediction of commercial-scale behaviour and not a measurement of it, and Stage 2 confirms it at scale. Every parameter in the simulation is drawn inside its normal operating range, which for this step means loads up to 105 L/m² and not the wider range explored in the characterization. The simulation propagates the fitted models, so it inherits their limitations, including the modest predicted R^2 of the MVM model.

Neither capability figure belongs to this step alone. Table 8.2 gives the modular clearance claim for the nominal batch, step by step, and the simulated mean in Table 8.1 is the distribution of that same claim across the 2,000 simulated batches.

Table 8.2: Modular viral clearance across the drug-substance process, by step ($\log_{10}$).

Step	XMuLV	MVM
Low-pH Viral Inactivation	7.10	0.00
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.87	10.03

Filtration supplies 5.32 log of the 10.03 log cumulative parvovirus claim, and anion exchange supplies 4.71 log of it (PCR-008). The low-pH hold contributes nothing to parvovirus clearance, because a non-enveloped virus is not inactivated by low pH, and it is not credited for it. For enveloped virus the picture is different. Filtration supplies 5.82 log of the 18.87 log cumulative claim, the low-pH hold supplies 7.10 log (PCR-006), and anion exchange supplies the rest. Protein A and cation exchange remove virus as well, and neither is credited in the claim. The consolidated position is in PCMR-001.

Robustness across the operating region rests on two findings from §5. Neither enveloped-virus clearance nor step yield varied significantly with load or with pressure over the ranges studied, so the only response that moves with an operating parameter is the one the step is designed to deliver. Both parameters are also easy to hold in practice. The volumetric load is fixed when the filter area is selected for the batch, and the pressure is set at the start of the filtration and monitored throughout under SOP-2012. The margins on both viral-clearance attributes are wide, and on that basis the small-virus safety of A-Mab drug substance is assured at commercial scale.

9 Parameter classification

Both parameters of the step are classified as well-controlled critical process parameters under SOP-4001. Each is linked to a critical quality attribute, and each has a low risk of leaving the design space in routine operation, which is the distinction the A-Mab framework draws between a CPP and a WC-CPP (CMC Biotech Working Group 2009). The classifications are recorded in Table 4.1 and the reasoning for each is given below.

- **Filtration volume (load), WC-CPP.** It has a demonstrated and quantified effect on MVM clearance, which is a very high criticality attribute, so it cannot be a general process parameter. It is not a CPP because the risk of leaving the design space is low. The load is fixed when the filter area is chosen for a batch and does not drift during operation, and its normal operating range sits well inside a characterized range over which the acceptance criterion is met everywhere (§7).
- **Filtration pressure, WC-CPP.** No effect of pressure on any response reached significance over 8–30 psi, and that null result is the evidence for the class, not an absence of evidence. Pressure is retained as a well-controlled critical process parameter, because it is the parameter that keeps the filter inside the operating window in which its retention has been established, and because the null result rests on a design with limited residual degrees of freedom. It is easy to control and it is monitored continuously, so it carries no risk of leaving the design space unnoticed.

The step has no critical process parameter, and it has no key process parameter. There is no parameter here whose control capability is poor relative to the width of its design space, which is the condition that would force the CPP designation. A key process parameter would drive process consistency without touching quality, and the step has no such parameter either. Step yield at this step is high and insensitive, and it is monitored as a process attribute.

The null pressure result is kept in the knowledge space. It is the evidence that the step tolerates the full operating window of the filter, and the control strategy in §10 relies on it when it assesses a pressure excursion against the design space.

10 Contribution to the control strategy

The step contributes four things to the control strategy of the drug substance process. Each follows from a result reported above, and each is carried into the campaign-level strategy in PCMR-001.

The first is a pair of controlled ranges. Volumetric load is controlled to 50–105 L/m² and filtration pressure to 10–20 psi, both inside the design space established in §6. The load range is the one that matters for quality, and its upper limit is the operative control on parvovirus clearance at this step.

The second is in-process monitoring. Filtration pressure is recorded throughout the filtration, and the volume filtered per unit membrane area is recorded and reconciled against the batch volume, both under SOP-2012. An excursion in either is assessed against the design space and against the proven acceptable ranges in §7. The pump ramp profile is now specified in SOP-2012 as well, following the deviation in §13.1, so that the set-point is reached without an overshoot. Neither enveloped-virus clearance nor step yield is monitored as a control point, because neither varied with load or with pressure over the ranges studied (§5.3).

The third is the filter itself. Every filter is flushed and pressure-tested before use and integrity-tested after use under SOP-2012, and a failed post-use integrity test invalidates the batch. Membrane lots are controlled by the vendor specification and by incoming verification, and the deviation in §13.2 led to the pre-use flux verification being retained as a routine record.

The fourth is the clearance credit itself. The log reduction this step delivers is claimed as a module of the cumulative viral clearance under ICH Q5A(R2) (International Council for Harmonisation 2023a), and it is claimed on the basis that its mechanism is independent of the mechanisms credited to the other steps. That independence is an argument about mechanism and not a measurement, and it is made in PCMR-001 where the whole claim is assembled.

The step does not control viral safety by itself, and nothing in this report should be read that way. Parvovirus clearance is shared with anion exchange (PCR-008) and enveloped-virus clearance with the low-pH hold (PCR-006) and anion exchange. Nor does the step control the quality of the material it receives. The composition of the anion exchange flow-through pool is set by the design space of Step 8, and a load outside that composition is outside what was characterized here.

11 Discussion

The step is well understood, and the understanding is concentrated in one relationship. MVM clearance declines with volumetric load, the decline is quantified across a range that extends well past the routine ceiling, and the required contribution is met at every point of that range including its worst corner. Everything else the study measured was flat. That is a simple result, and its simplicity is a property of the step. A size-based retention mechanism has few ways to fail, and the ones it has are covered by the post-use integrity test and by the load limit.

Confidence in the scale-down model is good for the operating-range conclusions and is necessarily weaker for the clearance figures themselves. Load per unit membrane area and pressure are the two quantities retention depends on, and both are matched between scales, so a range established on the model is a range for the commercial step. The clearance values are a different matter. They come from a spiked small-scale system that no commercial batch can reproduce, which is why ICH Q5A(R2) treats them as a modular claim supported by mechanism and not as a batch release measurement (International Council for Harmonisation 2023a). The margins reported in §6 and §8 should be read in that light.

Two deviations occurred during execution and both were dispositioned as retained. The pressure excursion reached 22 psi, above the normal operating range, and the membrane lot in the second deviation was 12 % below the vendor-typical flux. Neither changed a fitted effect, a classification or the operating region, and both are reported in full in §13. They are mentioned here because they are the only two places where the executed study departed from PCP-009.

Four limitations bound what this report establishes. The clearance figures rest on a small-scale spiking model, as noted above. The study used two model viruses, so nothing is established about the clearance of any other virus, and the parvovirus result should not be extended to other non-enveloped viruses without a mechanistic argument about size. The response-surface model has a modest predicted R^2 and rests on 12 runs, so it defines a region and predicts mean levels within it, and it does not support precise point predictions at the corners. Finally, the design space is not confirmed at scale at its edges, which is a general property of a characterization package and is addressed by the Stage 2 programme.

Each limitation is managed forward in a stated way. The spiking model and the choice of model viruses are the accepted approach of ICH Q5A(R2), and the claim they support is a modular one consolidated in PCMR-001. The precision of the model is managed by operating well inside the region, since the normal operating range for load stops far below the load at which the margin becomes small. Edge confirmation is a Stage 2 activity. No parameter at this step required the most stringent control, and the reasoning for that is in §9.

12 Conclusions

The small-virus retentive filtration step is well understood and robust across the ranges studied, and it delivers the clearance contribution the cumulative viral safety claim requires of it. Volumetric load governs parvovirus clearance and is the only parameter with a demonstrated effect on any response. Filtration pressure has no demonstrated effect over the full operating window of the filter, and enveloped-virus clearance and step yield are insensitive to both parameters over the ranges studied.

The design space is the characterized region of 50–140 L/m² of load and 8–30 psi of pressure. At its worst corner the fitted model predicts 4.48 log of MVM clearance against a required contribution of 3.89 log. Every proven acceptable range in §7 spans the whole characterization range under both analyses, so the binding constraint on the operating region is the multivariate corner and not any univariate range. Both parameters are classified as well-controlled critical process parameters, and the step has no critical process parameter and no key process parameter.

At commercial scale both governed attributes meet their acceptance criteria. Cumulative parvovirus clearance is the tighter capability at 1.51, with a mean 1.78 log above its floor. Two deviations were recorded and both were resolved without impact on the classifications or on the operating region. These conclusions are bounded by the characterization ranges, by the fitted response-surface model and by the qualified small-scale spiking model on which clearance was measured. They do not by themselves establish viral safety, which is a cumulative claim consolidated in PCMR-001 and confirmed at commercial scale in Stage 2.

13 Deviations from the plan

Two deviations from PCP-009 were recorded during execution of the study. Each was investigated, each was assessed for impact on the reported results, and each was dispositioned as retained, meaning that the affected runs remain in the analysis reported above. Table 13.1 summarizes them.

Table 13.1: Deviations recorded during execution of the study.

Deviation	Summary	Detected during	Disposition
DEV-009-01	Transient filtration-pressure excursion above NOR	in-process pressure monitoring	retained
DEV-009-02	Filter membrane lot flux below vendor-typical value	pre-use flux verification	retained

The two deviations are unrelated to each other. The first is an equipment event during a run and was found by in-process monitoring, and the second is a material attribute found before the affected runs began. Neither invalidated a design and neither required a study to be re-executed.

13.1 DEV-009-01 — transient filtration-pressure excursion above the normal operating range

Filtration pressure rose to 22 psi during one run, above the upper edge of the normal operating range of 10–20 psi. The excursion was transient and was detected by the in-process pressure record required by SOP-2012.

The investigation identified an overshoot on the pump ramp at the start of the filtration as the root cause. The pressure controller reached its set-point through an initial overshoot that decayed within the first part of the filtration, and the recorded maximum is that overshoot. No equipment fault was found, and the same behaviour was not seen on the other runs of the design.

The impact assessment rests on two facts. The excursion peak is inside the characterization range of 8–30 psi, at a coded position of +0.27, so the affected run sits inside the region the study explored. The fitted model predicts an MVM log reduction of 5.59 at that pressure and the set-point load, against a required step contribution of 3.89 log, and pressure was in any case shown in §5.3 to have no significant effect on any response. The deviation is therefore retained, the affected run remains in the analysis, and the classification of filtration pressure in §9 is unchanged. The pump ramp profile has been added to the operating parameters specified in SOP-2012 so that the overshoot is not repeated at commercial scale.

13.2 DEV-009-02 — filter membrane lot flux below the vendor-typical value

The membrane lot used for part of the study, LOT-VF-9331, showed a flux 12 % below the vendor-typical value at pre-use flux verification. A permeability deficit of that size raises a legitimate question about whether the lot is representative, and therefore about whether the retention characterized on it can be read as platform behaviour.

The investigation attributed the deficit to lot-to-lot variation in membrane permeability within the vendor specification. The lot met every release criterion of its certificate of analysis, including the vendor's own retention test, and the deficit is a shift within the specified permeability window.

The impact assessment rests on mechanism and on the reproducibility already reported. Flux is a permeability property of the membrane and is set by the open structure of the support, whereas retention is set by the pore-size distribution of the retentive layer, and the two are specified and released separately by the vendor. Any contribution the lot did make to the results is contained in the run-to-run scatter reported in §5.1, which was small enough to resolve the load effect without difficulty and a substantial part of which the method validation attributes to the infectivity assay (Appendix D). The MVM clearance of the affected runs was determined by AMV-3018 in the same way as every other run, and nothing in the diagnostics of Figure 5.2 marks those runs out. Every filter in the study passed its post-use integrity test.

The deviation is retained and the affected runs remain in the analysis. One change was made forward. Pre-use flux verification is now a routine record for the commercial step under SOP-2012, so that a lot at the edge of the vendor permeability window is identified before the batch is loaded.

14 Appendices A-D

The appendices carry the complete design matrices, the full effect and coefficient tables for every response, and the validated performance of the analytical methods. They are the primary record of the data summarized in §5.

14.1 Appendix A — Screening design matrix and responses

The coded screening design and the measured responses are given in Table 14.1. The coded settings of −1 / +1 mark the edges of the characterization range of each factor, and 0 marks its midpoint.

Table 14.1: Screening design matrix in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF (log_{10})	XMuLV LRF (log_{10})	Step yield
1	factorial	-1	-1	6.34	6.33	0.985
2	factorial	-1	1	6.08	6.41	0.972
3	factorial	1	-1	4.71	6.06	0.972
4	factorial	1	1	4.48	6.51	0.996
5	center	0	0	5.33	6.46	0.985
6	center	0	0	5.01	6.17	0.994
7	center	0	0	5.97	6.28	0.979

14.2 Appendix B — Response-surface design matrix and responses

The coded face-centred central composite design and the measured responses are given in Table 14.2.

Table 14.2: Response-surface design matrix in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF (log_{10})	XMuLV LRF (log_{10})	Step yield
1	factorial	-1	-1	6.23	6.2	0.989
2	factorial	-1	1	6	6.34	0.985

Run	Type	A	B	MVM LRF ($\log_{10}$)	XMuLV LRF ($\log_{10}$)	Step yield
3	factorial	1	-1	4.69	5.9	0.977
4	factorial	1	1	4.57	6.4	0.981
5	axial	-1	0	6.74	6.25	0.981
6	axial	1	0	4.51	6.38	0.979
7	axial	0	-1	5.11	6.15	0.975
8	axial	0	1	5.27	6.28	0.985
9	center	0	0	5.48	5.94	0.99
10	center	0	0	5.56	6.29	0.989
11	center	0	0	5.69	6.03	0.972
12	center	0	0	5.42	6.27	0.982

14.3 Appendix C — Full effect and coefficient tables

The screening effect table for step yield, which is referred to in §5.2, is given in Table 14.3.

Table 14.3: Screening effect estimates for step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AB	0.0181	0.00905	0.00365	2.48	0.0892	
A	0.00567	0.00283	0.00365	0.776	0.494	
B	0.00539	0.00269	0.00365	0.738	0.514	

The response-surface coefficient tables for the two responses not shown in §5.3 are given in Table 14.4 and Table 14.5, with their analyses of variance in Table 14.6 and Table 14.7. Neither model is significant overall, so neither table supports a predictive claim.

Table 14.4: Response-surface coefficients for XMuLV log reduction (coded factors).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.16	0.0719	85.7	1.7e-10	***
A	-0.0191	0.0643	-0.297	0.776	
B	0.128	0.0643	1.99	0.0943	
AB	0.0896	0.0788	1.14	0.299	
A^2	0.0849	0.0965	0.88	0.413	
B^2	-0.00707	0.0965	-0.0733	0.944	

Table 14.5: Response-surface coefficients for step yield (coded factors).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.982	0.00313	314	7.02e-14	***
A	-0.00336	0.0028	-1.2	0.275	
B	0.0016	0.0028	0.573	0.587	
AB	0.00199	0.00342	0.58	0.583	
A^2	-0.000294	0.00419	-0.0701	0.946	
B^2	0.000228	0.00419	0.0544	0.958	

Table 14.6: Analysis of variance with lack-of-fit partition, XMuLV log reduction.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.153	5	0.0306	1.23	0.398
Residual	0.149	6	0.0248	nan	nan
Lack of fit	0.0562	3	0.0187	0.606	0.655
Pure error	0.0927	3	0.0309	nan	nan

Table 14.7: Analysis of variance with lack-of-fit partition, step yield.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.9e-05	5	1.98e-05	0.422	0.819
Residual	0.000282	6	4.69e-05	nan	nan
Lack of fit	7.02e-05	3	2.34e-05	0.332	0.805
Pure error	0.000211	3	7.05e-05	nan	nan

14.4 Appendix D — Analytical methods summary

The validated performance of the two infectivity assays used in the study is given in Table 14.8. Precision for both is reported as a log10 standard deviation, which is the natural unit for a titre-based method, and the variance share is the fraction of observed variability the validation attributes to the method.

Table 14.8: Validated performance of the analytical methods used at the step.

Method	Title	Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32 log10 SD	1.4 log10 TCID50/mL	96	0.3
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3 log10 SD	1.5 log10 TCID50/mL	96	0.28

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